

JEE Mains 2019 Chapter wise Question Bank

Continuity and Differentiability – Questions

Q1

Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a function defined as

$$f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a + bx, & \text{if } 1 < x < 3 \\ b + 5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$$

Then, f is :

- (1) continuous if $a = 5$ and $b = 5$
- (2) continuous if $a = -5$ and $b = 10$
- (3) continuous if $a = 0$ and $b = 5$
- (4) not continuous for any values of a and b

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Q2

$$\text{Let } f(x) = \begin{cases} \max\{|x|, x^2\} & |x| \leq 2 \\ 8 - 2|x|, & 2 < |x| \leq 4 \end{cases}$$

Let S be the set of points in the interval $(-4, 4)$ at which f is not differentiable. Then S :

- (1) is an empty set
- (2) equals $\{-2, -1, 0, 1, 2\}$
- (3) equals $\{-2, -1, 1, 2\}$
- (4) equals $\{-2, 2\}$

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Q3

Let $f: (-1, 1) \rightarrow \mathbf{R}$ be a function defined by $f(x) = \max\{-|x|, -\sqrt{1-x^2}\}$. If K be the set of all points at which f is not differentiable, then K has exactly:

- (1) five elements
- (2) one element
- (3) three elements
- (4) two elements

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Q4

$$\text{Let } f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases} \text{ and}$$

 $g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is :

- (1) differentiable at all points
- (2) not continuous
- (3) not differentiable at two points
- (4) not differentiable at one point

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Q5

Let K be the set of all real values of x where the function $f(x) = \sin|x| - |x| + 2(x - \pi) \cos|x|$ is not differentiable.Then the set K is equal to :

- (1) ϕ (an empty set)
- (2) $\{\pi\}$
- (3) $\{0\}$
- (4) $\{0, \pi\}$

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Q6

Continuity and Differentiability

Let S be the set of all points in $(-\pi, \pi)$ at which the function $f(x) = \min \{\sin x, \cos x\}$ is not differentiable. Then S is a subset of which of the following?

- (1) $\left\{-\frac{\pi}{4}, 0, \frac{\pi}{4}\right\}$ (2) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4}\right\}$
(3) $\left\{-\frac{\pi}{2}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{\pi}{2}\right\}$ (4) $\left\{-\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{4}\right\}$

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Q7

Let $f: [-1, 3] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3, \end{cases}$$

where $[t]$ denotes the greatest integer less than or equal to t . Then, f is discontinuous at :

- (1) only one point (2) only two points
(3) only three points (4) four or more points

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Q8

If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by

$$f(x) = \begin{cases} \frac{\sqrt{2} \cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$$

is continuous, then k is equal to:

- (1) 2 (2) $\frac{1}{2}$ (3) 1 (4) $\frac{1}{\sqrt{2}}$

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Q9

Let $f(x) = 15 - |x - 10|$; $x \in \mathbb{R}$. Then the set of all values of x , at which the function, $g(x) = f(f(x))$ is not differentiable, is:

- (1) $\{5, 10, 15\}$ (2) $\{10, 15\}$
(3) $\{5, 10, 15, 20\}$ (4) $\{10\}$

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Q10

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If $f(x) = [x] - \left[\frac{x}{4}\right]$, $x \in \mathbb{R}$, where $[x]$ denotes the greatest integer function, then:

- (1) f is continuous at $x = 4$.
(2) $\lim_{x \rightarrow 4+} f(x)$ exists but $\lim_{x \rightarrow 4-} f(x)$ does not exist.
(3) Both $\lim_{x \rightarrow 4-} f(x)$ and $\lim_{x \rightarrow 4+} f(x)$ exist but are not equal.
(4) $\lim_{x \rightarrow 4-} f(x)$ exists but $\lim_{x \rightarrow 4+} f(x)$ does not exist.

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Q11

If the function

$$f(x) = \begin{cases} a|\pi - x| + 1, & x \leq 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$$

is continuous at $x = 5$, then the value of $a - b$ is:

- (1) $\frac{2}{\pi + 5}$ (2) $\frac{-2}{\pi + 5}$ (3) $\frac{2}{\pi - 5}$ (4) $\frac{2}{5 - \pi}$

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Q12

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable at $c \in \mathbb{R}$ and $f(c) = 0$. If $g(x) = |f(x)|$, then at $x = c$, g is :

- (1) not differentiable if $f'(c) = 0$
(2) differentiable if $f'(c) \neq 0$
(3) differentiable if $f'(c) = 0$
(4) not differentiable

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Q13

$$\text{If } f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x}, & x < 0 \\ q, & x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{x^{3/2}}, & x > 0 \end{cases}$$

is continuous at $x = 0$, then the ordered pair (p, q) is equal to:

- (1) $\left(-\frac{3}{2}, -\frac{1}{2}\right)$ (2) $\left(-\frac{1}{2}, \frac{3}{2}\right)$
(3) $\left(-\frac{3}{2}, \frac{1}{2}\right)$ (4) $\left(\frac{5}{2}, \frac{1}{2}\right)$

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Q14

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function such that $f(2) = 6$ and $f'(2) = \frac{1}{48}$.

If $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$, then $\lim_{x \rightarrow 2} g(x)$ is equal to :

- (1) 18 (2) 24 (3) 12 (4) 36

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Continuity and Differentiability - Solutions

Q1

(4) Let $f(x)$ is continuous at $x = 1$, then

$$f(1^-) = f(1) = f(1^+)$$

$$\Rightarrow 5 = a + b \quad \dots(1)$$

Let $f(x)$ is continuous at $x = 3$, then

$$f(3^-) = f(3) = f(3^+)$$

$$\Rightarrow a + 3b = b + 15 \quad \dots(2)$$

Let $f(x)$ is continuous at $x = 5$, then

$$f(5^-) = f(5) = f(5^+)$$

$$\Rightarrow b + 25 = 30$$

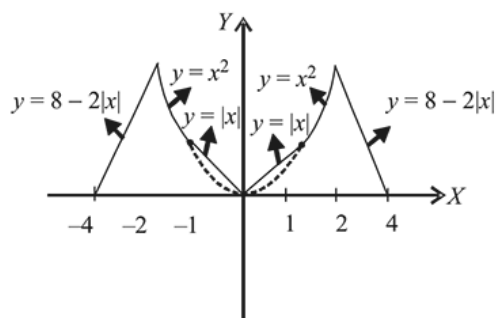
$$\Rightarrow b = 30 - 25 = 5$$

From (1), $a = 0$ But $a = 0$, $b = 5$ do not satisfy equation (2)Hence, $f(x)$ is not continuous for any values of a and b

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Q2

$$(2) \text{ Given } f(x) = \begin{cases} \max\{|x|, x^2\} & |x| \leq 2 \\ 8 - 2|x| & 2 < |x| \leq 4 \end{cases}$$

 $\therefore f(x)$ is not differentiable at $-2, -1, 0, 1$ and 2 .

$$\therefore S = \{-2, -1, 0, 1, 2\}$$

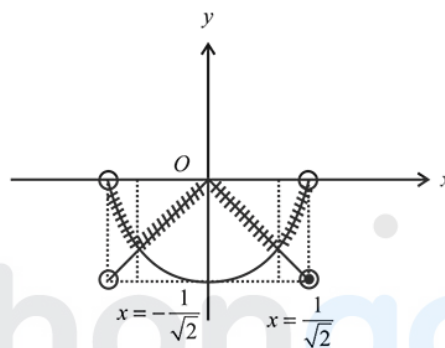
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Q3

(3) Consider the function

$$f(x) = \max\{-|x|, -\sqrt{1-x^2}\}$$

Now, the graph of the function

From the graph, it is clear that $f(x)$ is not differentiable at $x = 0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

$$\text{Then, } K = \left\{-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right\}$$

Hence, K has exactly three elements.

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Q4

Continuity and Differentiability

$$(4) f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$$

$$\text{Then, } f(|x|) = \begin{cases} -1, & -2 \leq |x| < 0 \\ |x|^2 - 1, & 0 \leq |x| \leq 2 \end{cases}$$

$$\Rightarrow f(|x|) = x^2 - 1, -2 \leq x \leq 2$$

$$\Rightarrow g(x) = \begin{cases} 1 + x^2 - 1, & -2 \leq x < 0 \\ (x^2 - 1) + |x^2 - 1|, & 0 \leq x \leq 2 \end{cases}$$

$$= \begin{cases} x^2, & -2 \leq x < 0 \\ 0, & 0 \leq x < 1 \\ 2(x^2 - 1), & 1 \leq x \leq 2 \end{cases}$$

$$g'(0^-) = 0, g'(0^+) = 0, g'(1^-) = 0, g'(1^+) = 4$$

$$\Rightarrow g(x) \text{ is non-differentiable at } x = 1$$

$$\Rightarrow g(x) \text{ is not differentiable at one point.}$$

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Q5

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$$(1) f(x) = \sin |x| - |x| + 2(x - \pi) \cos |x|$$

There are two cases,

Case (1), $x > 0$

$$f(x) = \sin x - x + 2(x - \pi) \cos x$$

$$f'(x) = \cos x - 1 + 2(1 - 0) \cos x - 2 \sin(x - \pi)$$

$$f'(x) = 3 \cos x - 2(x - \pi) \sin x - 1$$

Then, function $f(x)$ is differentiable for all $x > 0$

Case (2) $x < 0$

$$f(x) = -\sin x + x + 2(x - \pi) \cos x$$

$$f'(x) = -\cos x + 1 - 2(x - \pi) \sin x + 2 \cos x$$

$$f'(x) = \cos x + 1 - 2(x - \pi) \sin x$$

Then, function $f(x)$ is differentiable for all $x < 0$

Now check for $x = 0$

$$f'(0^+) \text{ R.H.D.} = 3 - 1 = 2$$

$$f'(0^-) \text{ L.H.D.} = 1 + 1 = 2$$

$$\text{L.H.D.} = \text{R.H.D.}$$

Then, function $f(x)$ is differentiable for $x = 0$. So it is differentiable everywhere

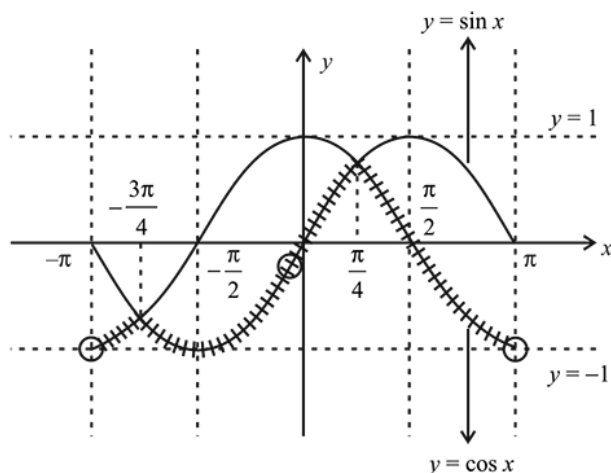
$$\text{Hence, } k = \phi$$

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Q6

Continuity and Differentiability

(2) $f(x) = \min\{\sin x, \cos x\}$



$\therefore f(x)$ is not differentiable at $x = -\frac{3\pi}{4}, \frac{\pi}{4}$

$$\therefore S = \left\{-\frac{3\pi}{4}, \frac{\pi}{4}\right\}$$

$$\Rightarrow S \subseteq \left\{-\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4}\right\}$$

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Q7

(3) Given function is,

$$f(x) = \begin{cases} |x| + [x], & -1 \leq x < 1 \\ x + |x|, & 1 \leq x < 2 \\ x + [x], & 2 \leq x \leq 3 \end{cases}$$

$$= \begin{cases} -x-1, & -1 \leq x < 0 \\ x, & 0 \leq x < 1 \\ 2x, & 1 \leq x < 2 \\ x+2, & 2 \leq x < 3 \\ 6, & x = 3 \end{cases}$$

$$\Rightarrow f(-1) = 0, f(-1^+) = 0;$$

$$f(0^-) = -1, f(0) = 0, f(0^+) = 0;$$

$$f(1^-) = 1, f(1) = 2, f(1^+) = 2;$$

$$f(2^-) = 4, f(2) = 4, f(2^+) = 4;$$

$$f(3^-) = 5, f(3) = 6$$

$f(x)$ is discontinuous at $x = \{0, 1, 3\}$

Hence, $f(x)$ is discontinuous at only three points.

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Q8

(2) Since, $f(x)$ is continuous, then

$$\lim_{x \rightarrow \frac{\pi}{4}} f(x) = f\left(\frac{\pi}{4}\right)$$

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} \cos x - 1}{\cot x - 1} = k$$

Now by L- hospital's rule

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} \sin x}{\operatorname{cosec}^2 x} = k \Rightarrow \frac{\sqrt{2} \left(\frac{1}{\sqrt{2}}\right)}{(\sqrt{2})^2} = k \Rightarrow k = \frac{1}{2}$$

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Q9

(1) Since, $f(x) = 15 - |(10 - x)|$

$$\therefore g(x) = f(f(x)) = 15 - |10 - [15 - |10 - x||]$$

$$= 15 - ||10 - x| - 5|$$

$\therefore$ Then, the points where function $g(x)$ is Non-differentiable are

$$10 - x = 0 \text{ and } |10 - x| = 5$$

$$\Rightarrow x = 10 \text{ and } x - 10 = \pm 5$$

$$\Rightarrow x = 10 \text{ and } x = 15, 5$$

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Q10

$$(1) \text{ L.H.L. } \lim_{x \rightarrow 4^-} \left([x] - \left[\frac{x}{4} \right] \right) = 3 - 0 = 3$$

$$\text{R.H.L. } \lim_{x \rightarrow 4^+} \left([x] - \left[\frac{x}{4} \right] \right) = 4 - 1 = 3$$

$$f(4) = [4] - \left[\frac{4}{4} \right] = 4 - 1 = 3$$

$$\therefore \text{LHL} = f(4) = \text{RHL}$$

$$\therefore f(x) \text{ is continuous at } x = 4$$

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Q11

Continuity and Differentiability

$$(4) \text{ R.H.L. } \lim_{x \rightarrow 5^+} b|(x - \pi)| + 3 = (5 - \pi)b + 3$$

$$f(5) = \text{L.H.L. } \lim_{x \rightarrow 5^-} a|(\pi - x)| + 1 = a(5 - \pi) + 1$$

$\therefore$ function is continuous at $x = 5$

$\therefore$ LHL = RHL

$$(5 - \pi)b + 3 = (5 - \pi)a + 1$$

$$\Rightarrow 2 = (a - b)(5 - \pi) \Rightarrow a - b = \frac{2}{5 - \pi}$$

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Q12

$$(3) g'(c) = \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c}$$

$$\Rightarrow g'(c) = \lim_{x \rightarrow c} \frac{|f(x)| - |f(c)|}{x - c}$$

Since, $f(c) = 0$

$$\text{Then, } g'(c) = \lim_{x \rightarrow c} \frac{|f(x)|}{x - c}$$

$$\Rightarrow g'(c) = \lim_{x \rightarrow c} \frac{f(x)}{x - c}; \text{ if } f(x) > 0$$

$$\text{and } g'(c) = \lim_{x \rightarrow c} \frac{-f(x)}{x - c}; \text{ if } f(x) < 0$$

$$\Rightarrow g'(c) = f'(c) = -f'(c)$$

$$\Rightarrow 2f'(c) = 0 \Rightarrow f'(c) = 0$$

Hence, $g(x)$ is differentiable if $f'(c) = 0$

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Q13

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$$(3) f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & x < 0 \\ q & x = 0 \\ \frac{\sqrt{x^2 + x} - \sqrt{x}}{\frac{3}{x^2}} & x > 0 \end{cases} \text{ is continuous at } x = 0$$

Therefore, $f(0^-) = f(0) = f(0^+)$... (1)

$$f(0^-) = \lim_{h \rightarrow 0} f(0 - h) = \lim_{h \rightarrow 0} \frac{\sin(p+1)(-h) + \sin(-h)}{-h}$$

$$= \lim_{h \rightarrow 0} \left[\frac{-\sin(p+1)h}{-h} + \frac{\sin h}{h} \right]$$

$$= \lim_{h \rightarrow 0} \frac{\sin(p+1)h}{h(p+1)} \times (p+1) + \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$= (p+1) + 1 = p+2 \quad \dots (2)$$

$$\text{And } f(0^+) = \lim_{h \rightarrow 0} f(0 + h) = \frac{\sqrt{h^2 + h} - \sqrt{h}}{h^{3/2}}$$

$$= \lim_{h \rightarrow 0} \frac{(h)^{\frac{1}{2}} [\sqrt{h+1} - 1]}{h \left(\frac{1}{h^{\frac{1}{2}}} \right)}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h+1} - 1}{h} \times \frac{\sqrt{h+1} + 1}{\sqrt{h+1} + 1} = \lim_{h \rightarrow 0} \frac{h+1-1}{h(\sqrt{h+1} + 1)}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+1} + 1} = \frac{1}{1+1} = \frac{1}{2} \quad \dots (3)$$

Now, from equation (1),

$$f(0^-) = f(0) = f(0^+) \Rightarrow p+2 = q = \frac{1}{2}$$

$$\Rightarrow q = \frac{1}{2} \text{ and } p = \frac{-3}{2} \therefore (p, q) \left(-\frac{3}{2}, \frac{1}{2} \right)$$

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Q14

(1) Given, $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$

Differentiating both sides,

$$4(f(x))^3 \cdot f'(x) = g'(x)(x-2) + g(x)$$

Putting $x = 2$, $\frac{4(6)^3 \cdot 1}{48} = g(2) \Rightarrow \lim_{x \rightarrow 2} g(x) = 18$

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